VICTORY 2027

The Circulatory System

LECTURE-6 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. The Human Heart


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

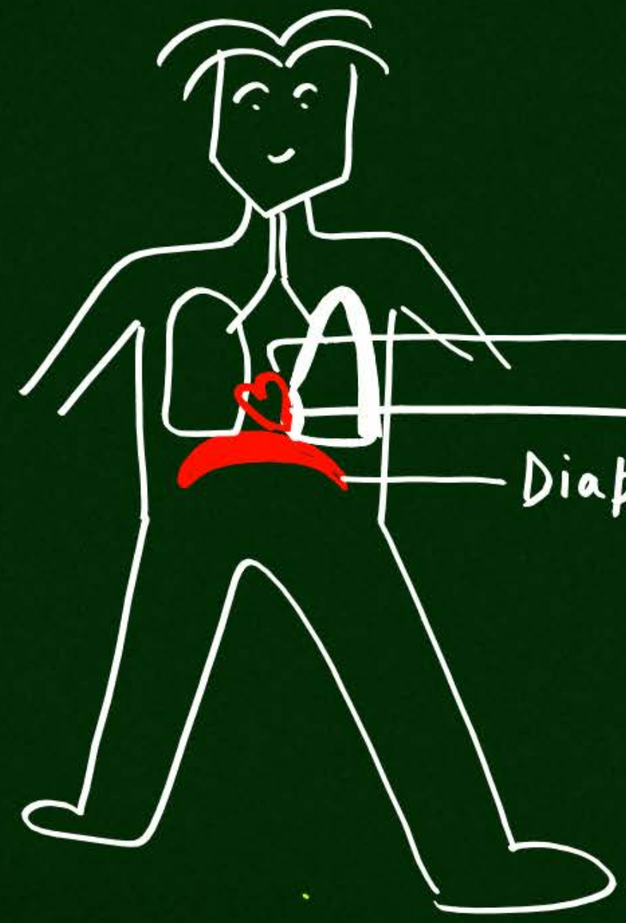
RATTON KO KARO **BYE BYE !!**

EASY CONCEPTS SE KARO **HELLO HI !!**

With – *Uppu Ma'am*



Human Heart



Thoracic cavity

Abdominal cavity

[MEDIASTINUM] cavity b/w lungs where ♡ is located.

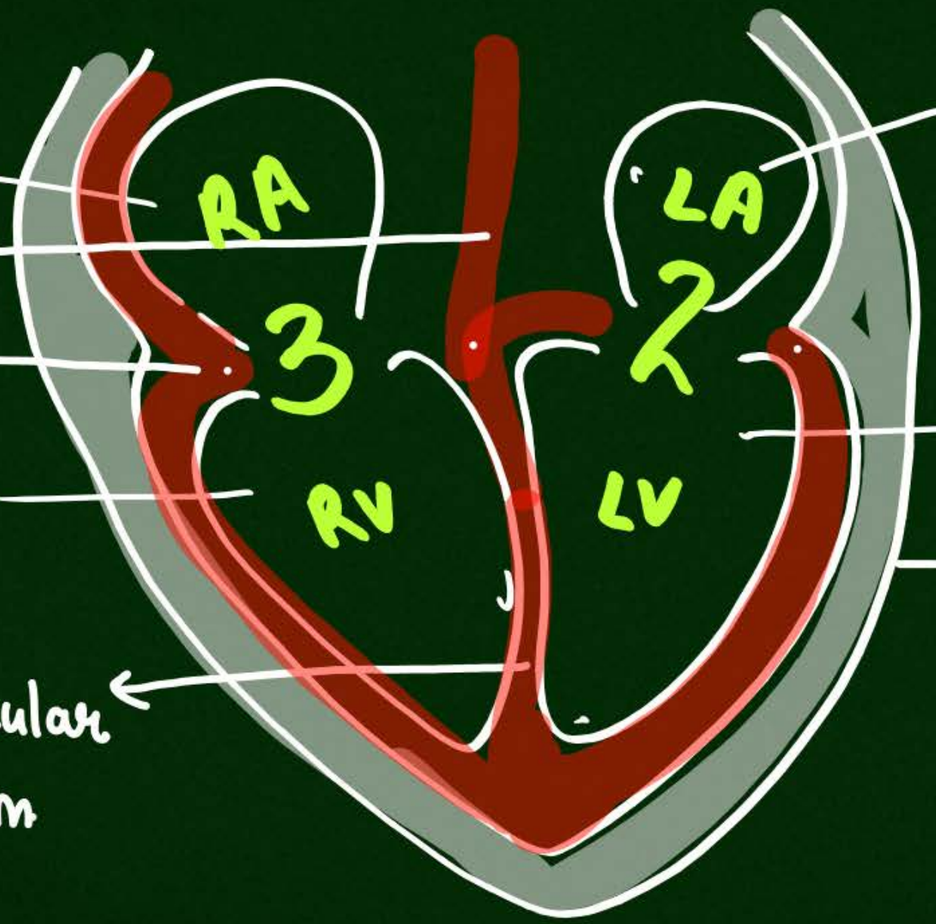
Diaphragm

Cardia Notch

→ the depression in the left lung for the placement of heart.

RIGHT

LEFT



Right Atrium

I. Atrial Septum

I. Atrial Ventricular Septum

R. Ventricle.

Inter Ventricular Septum

Left Atrium / Auricle

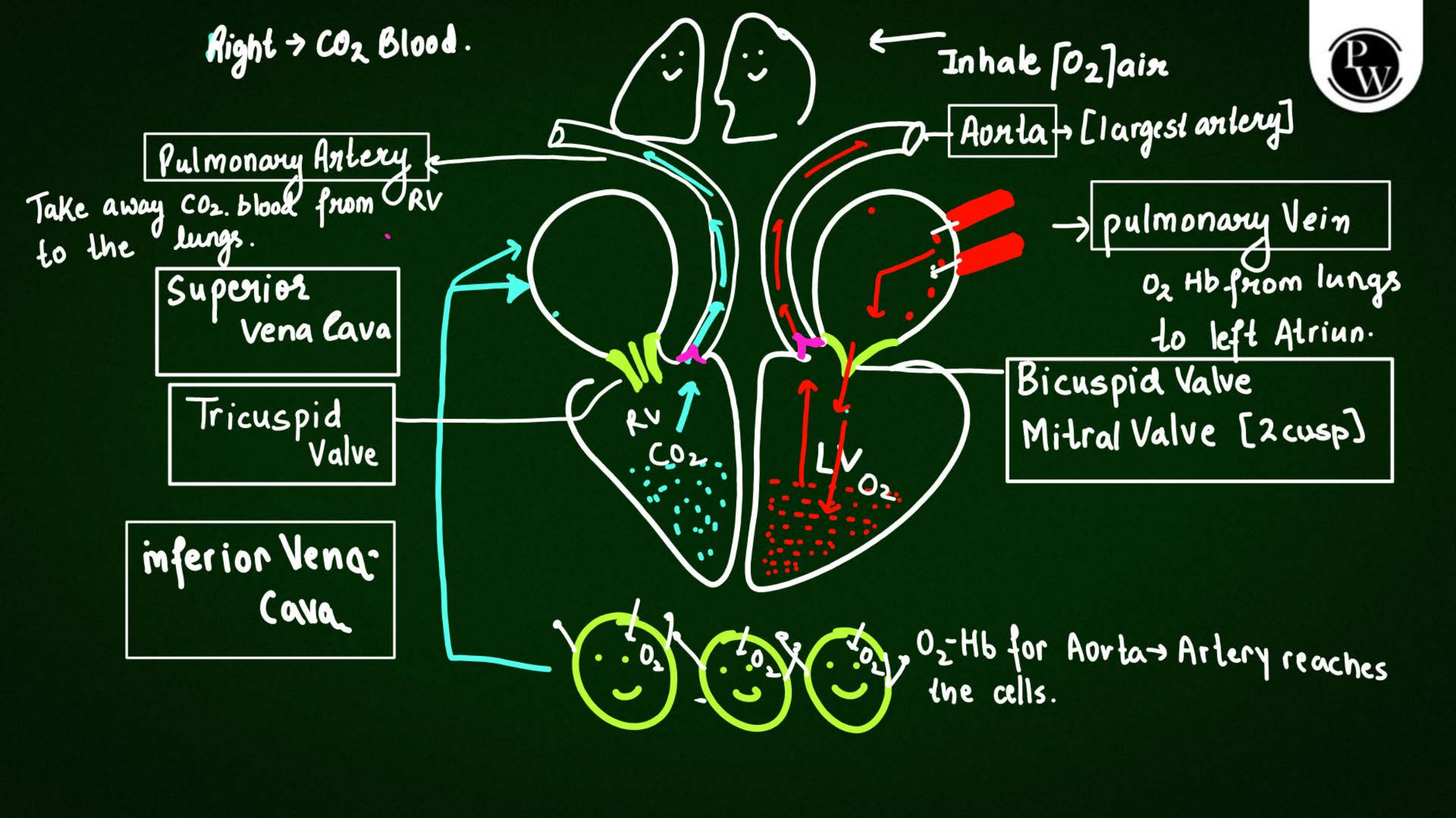
L. Ventricle.

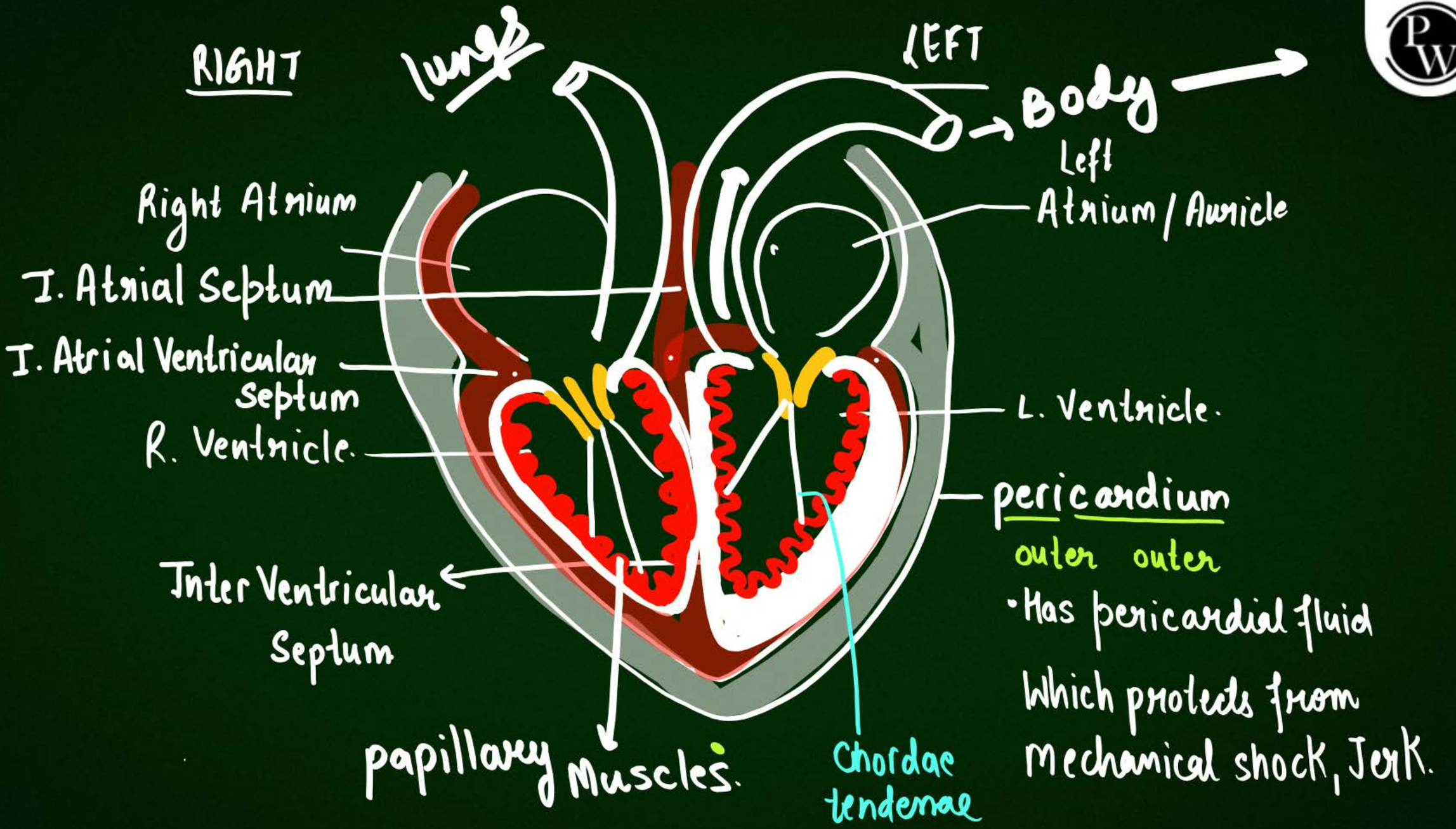
pericardium

outer outer

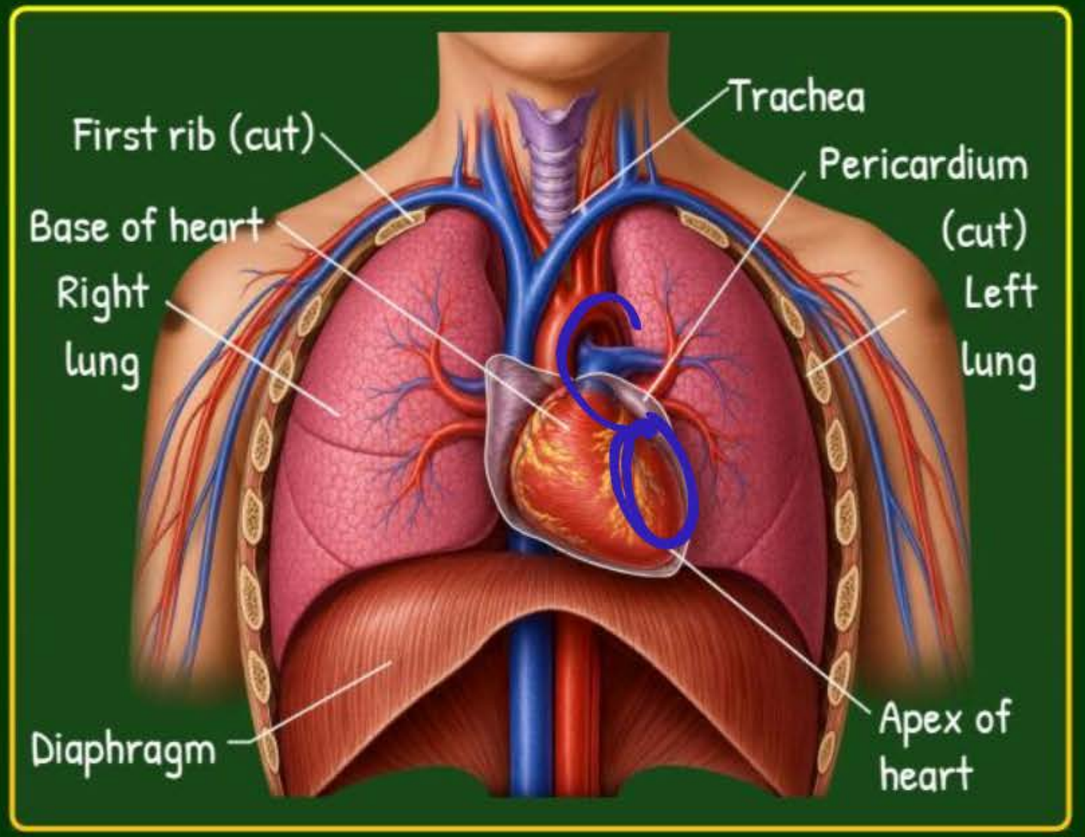
- Has pericardial fluid

Which protects from mechanical shock, Jerk.


Right → CO_2 Blood.
Inhale [O_2] air
Aorta → [largest artery]
Pulmonary Artery
Take away CO_2. blood from RV to the lungs.
pulmonary Vein
O_2 Hb from lungs to left Atriun.
Superior Vena Cava
Tricuspid Valve
Bicuspid Valve
Mitral Valve [2 cusp]
RV
CO_2
LV
O_2
inferior Vena Cava
O_2
O_2
O_2
O_2-Hb for Aorta → Artery reaches the cells.


RIGHT
Lungs
LEFT
Body
Left Atrium / Auricle
Right Atrium
I. Atrial Septum
I. Atrial Ventricular Septum
R. Ventricle.
L. Ventricle.
pericardium
outer outer
• Has pericardial fluid
Which protects from mechanical shock, Jerk.
Inter Ventricular Septum
papillary Muscles.
Chordae tendenae

Feature	Description
Location	In the centre between the lungs and above the diaphragm.
Dimensions	12 cm in length and 9 cm in width.
Size	In adult humans, it is about the size of one's fist.
Covering	Covered by a double membrane, the pericardium. It contains lubricating pericardial fluid, which protects the heart from mechanical injuries.



Q Heart is located in

A) In the centre b/w 2 lungs

B) Above diaphragm

C) In Mediastinum

D) All of the above



Q. The outer membrane of human heart is

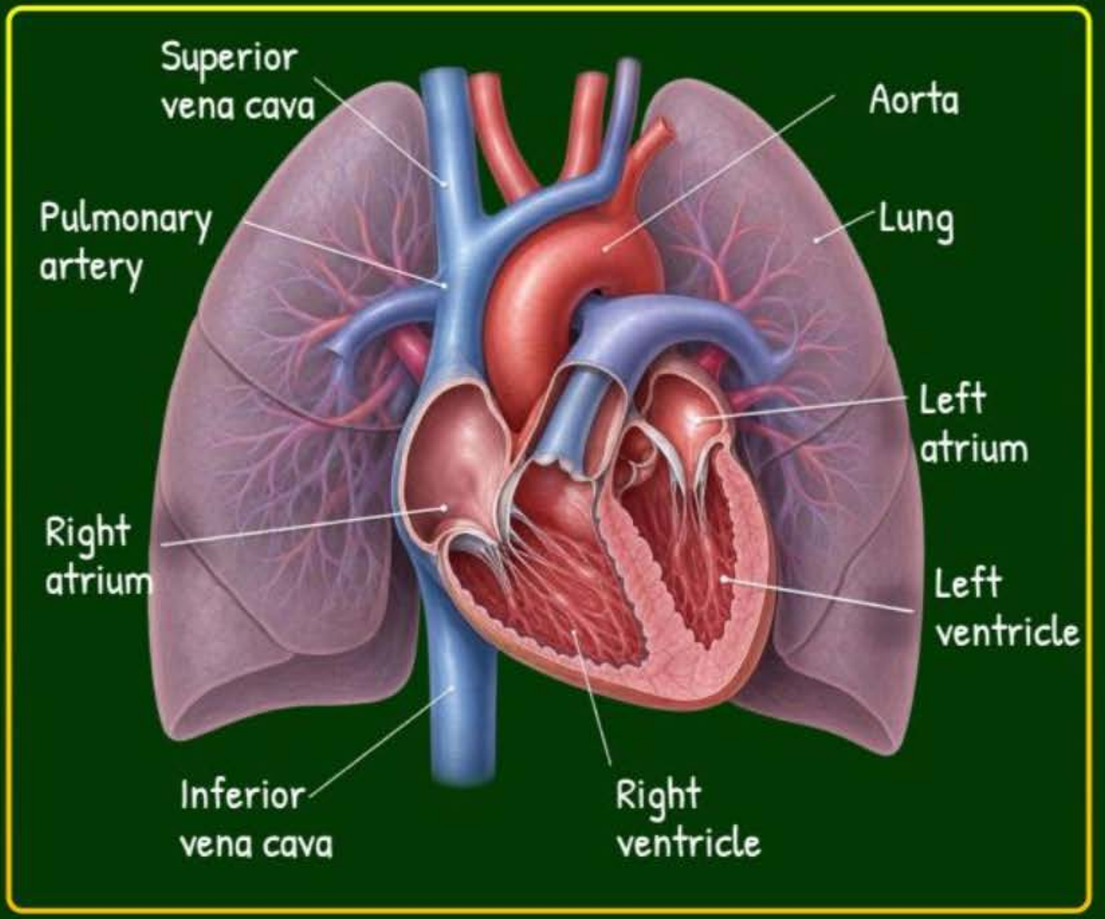
A) Pleura (lungs)

B) Pericardium

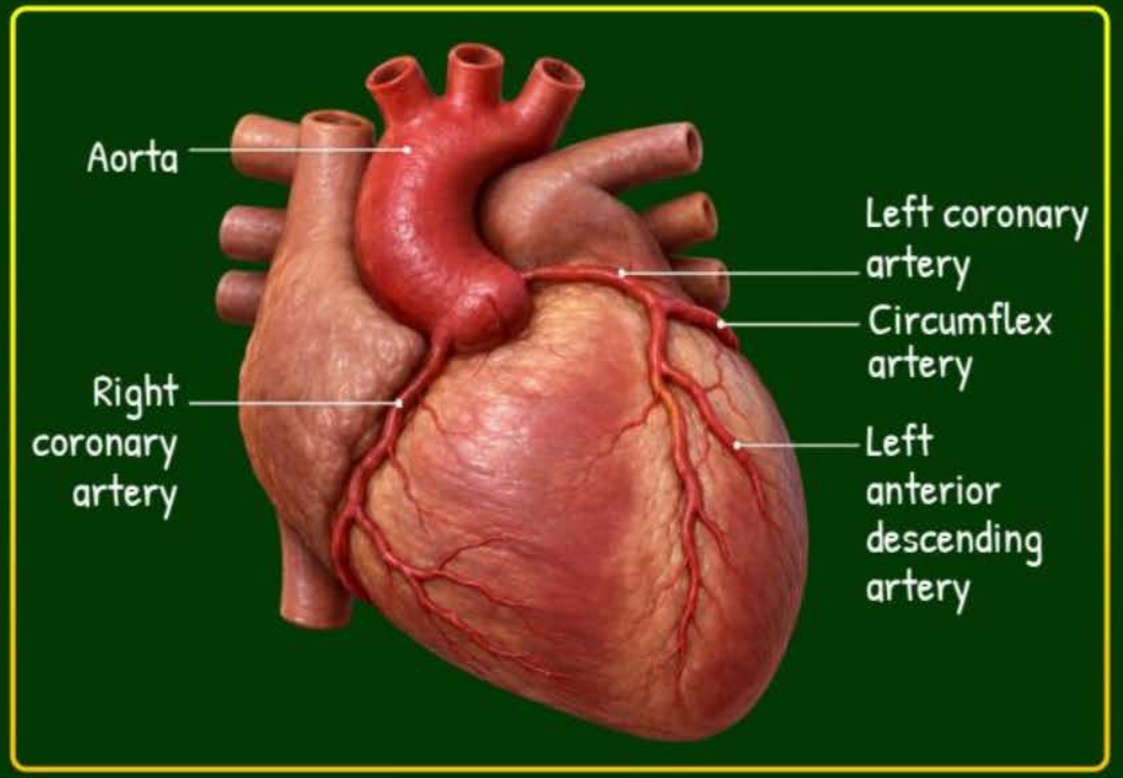
C) Meninges

D) None of the above

Topic	Details
Chambers of the Heart	Two upper atria and two lower ventricles.
Blood Vessels Entering the Heart	**Superior vena cava** (anterior vena cava/precaval) brings deoxygenated blood from the anterior part of the body, i.e., head, chest and arms.
Blood Vessels Entering the Heart	**Inferior vena cava** (posterior vena cava/postcaval) brings blood from the posterior region of the body, including abdomen and legs.
Blood Vessels Entering the Heart	**Pulmonary veins** carry oxygenated blood from the lungs to the left atrium.

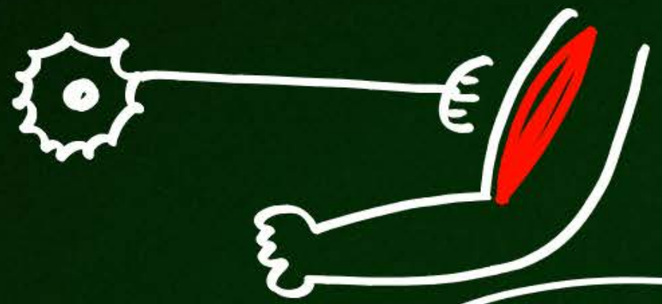


Topic	Details
Blood Vessels Leaving the Heart	The pulmonary artery arises from the right ventricle and carries deoxygenated blood to the lungs for oxygenation.
Blood Vessels Leaving the Heart	The aorta arises from the left ventricle and carries oxygenated blood to supply it to all parts of the body.
Coronary Arteries	Two coronary arteries rising from the base of the aorta supply blood to heart muscles, i.e., cardiac muscles.



Topic	Details
Heart Valves	Tricuspid valve (right atrio-ventricular valve) is located between the right atrium and the right ventricle.
	Bicuspid valve (left atrio-ventricular valve) is located between the left atrium and the left ventricle.
	Pulmonary semilunar valves are located at the opening of the right ventricle in the pulmonary artery.
	Aortic semilunar valves are located at the opening of the left ventricle in the aorta.

Muscles can contract & Relax by taking signal from Neuron



Cardiac Muscles are special and does not take signal from Neurons.

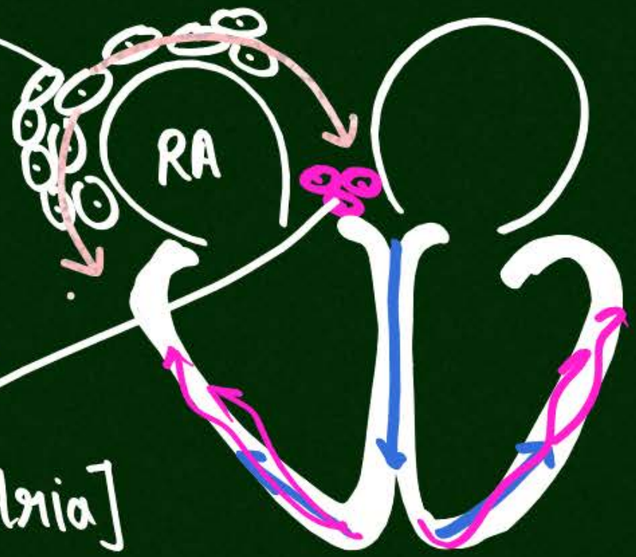
Sino Atrial Node [to the Right side of RA.]
These are cardiac muscles which generate the signal of contraction.

↓

Atrio Ventriculon Node. [B/w two atria]

↓

Bundle of His → Purkinje fibres.



QUESTIONS

The blood vessels carry blood from the heart to the various parts of the body are called _____.

A Arteries ✓

B Veins

C Septum

D More than one of the above

QUESTIONS

The total volume of blood in a normal adult human being is

A 5.6 litres

B 3.4 litres

C 8.10 litres

D More than one of the above

QUESTIONS

The opening of right atrium into right ventricle is guarded by _____.

A. Mitral valve

B. Tricuspid valve

C. Bicuspid valve

D. Aortic semilunar valve

RA → RV

QUESTIONS

Mitral valve is present between _____.

A. Right atrium and left ventricle

B. Right and left ventricle

C. Left ventricle and aorta

D. Left atrium and left ventricle ✓

Thank You